Power BI vs Excel, side by side

Each page of the ExportTest report as Power BI Desktop draws it, next to the sheet the export writes for it.
Excel pages are rendered by desktop Excel from the exported workbook; nothing is retouched.

Pages compared

PAGE	REPORT PAGE	EXCEL MODE	RESULT
2	**Colour test**	Excel charts	MATCHES
3	**Charts mode**	Excel charts	MATCHES
4	**All visuals 1 – Bars and columns**	Excel charts	MATCHES
5	**All visuals 2 – Lines and areas**	Excel charts	MATCHES
6	**All visuals 3 – Parts, flows and spread**	Excel charts	CLOSE
7	**All visuals 4 – Treemap and maps**	Data tables	AS A TABLE
8	**All visuals 5 – Cards and KPIs**	Data tables	MATCHES
9	**All visuals 6 – Tables and matrices**	Data tables	MATCHES
10	**All visuals 7 – Slicers, AI and other**	Data tables	MATCHES
11	**MS visuals 3 – Bullet, Gantt, Histogram, Dual KPI**	Data tables	MATCHES
12	**Double Donut**	Data tables	MATCHES
13	**Nested Bars**	Data tables	MATCHES
14	**Comparison Table**	Data tables	MATCHES

At a glance

11
pages that match

1
close, with named differences

1
shown as tables (no Excel equivalent)

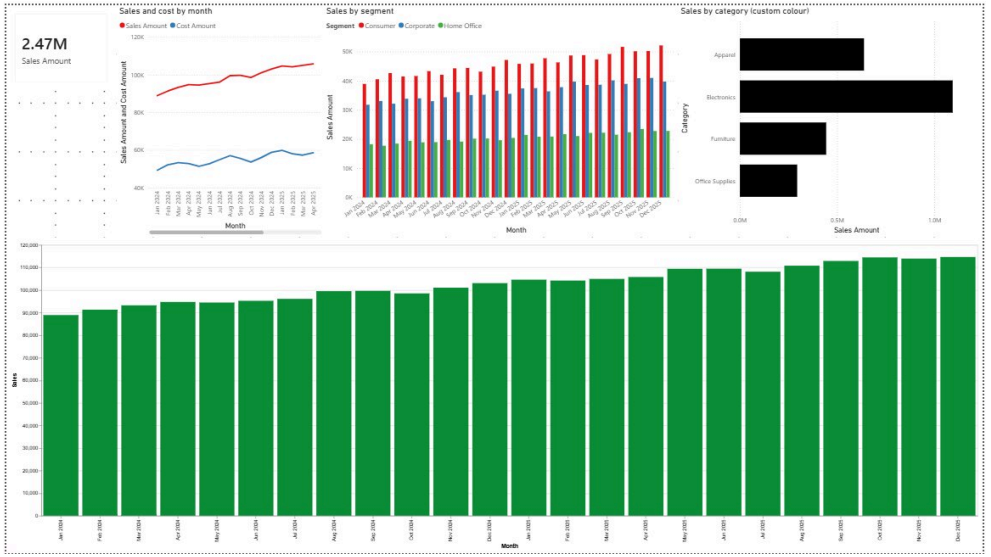
How it was produced

- Power BI side: ExportTest.pbip open in Power BI Desktop, each page captured as drawn.
- Excel side: the export run on the same report (both modes), opened in desktop Excel and printed page by page.
- Values, colours and order were compared visual by visual; every difference is named on its page.
- Charts pages use "Visuals as: Charts"; the other pages use "Data tables".

02 Colour test MATCHES

EXCEL MODE · EXCEL CHARTS

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

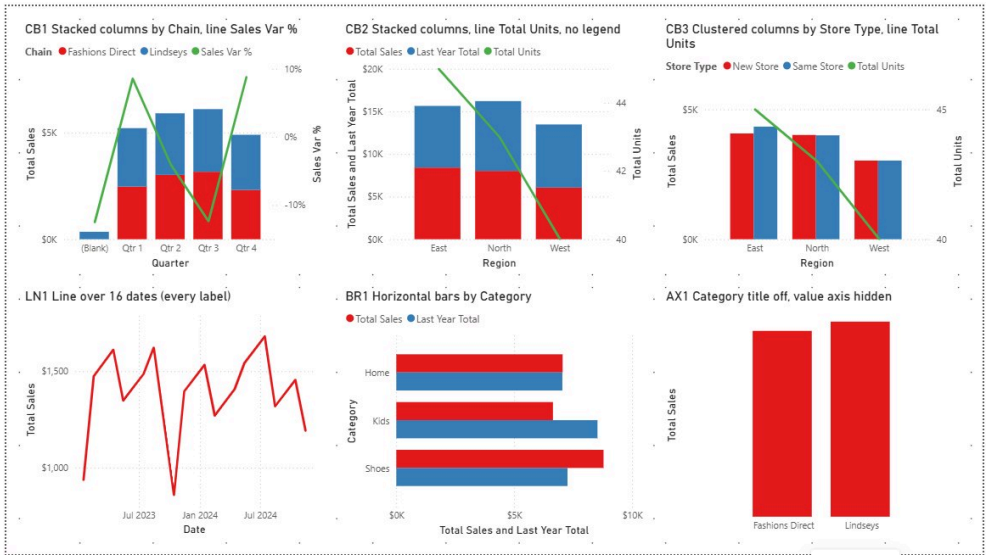
- Line and column charts in the report's own theme (red, blue, green), legends in the same order
- The bar chart's colour set by hand (black) carried over exactly
- The card becomes a KPI tile; values to the cent

DIFFERENCES

- Deneb (a custom Vega-Lite spec) is a table: its colour comes from the Bar Color field in the data

03 Charts mode MATCHES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

- Combo charts keep their line on a secondary axis, read once per category even with a legend
- Axis titles and hidden axes as in the report; every category label shown; bars start at zero, so a hidden axis never exaggerates a difference
- Horizontal bars, clustered and stacked columns in the series colours

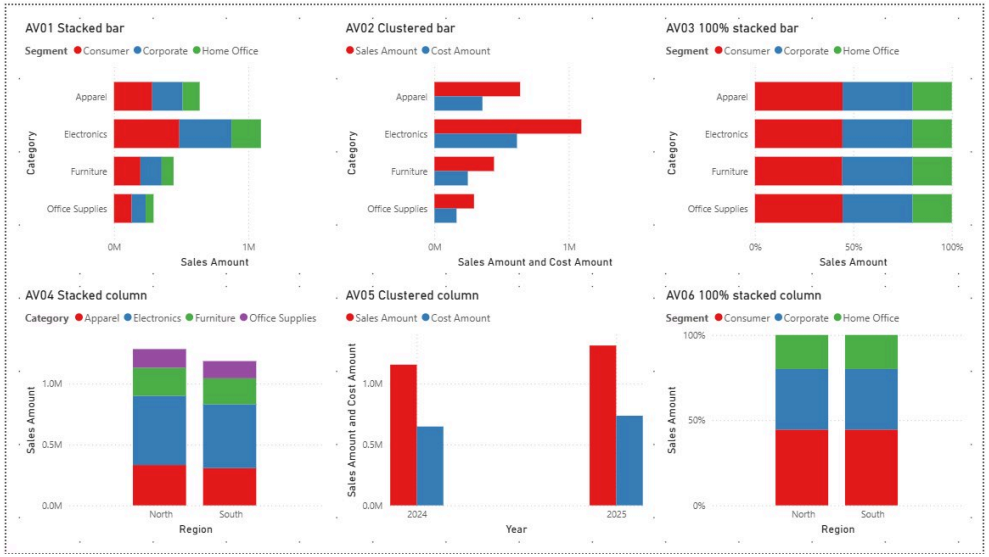
DIFFERENCES

- Excel picks its own axis steps where Power BI uses fewer gridlines

04 All visuals 1 – Bars and columns MATCHES

EXCEL MODE · EXCEL CHARTS

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

- All six bar and column types are native Excel charts: stacked, clustered and 100%
- Legend values and series in the theme colours, in Power BI's order
- Axis titles (Category, Sales Amount, Region, Year) as in the report

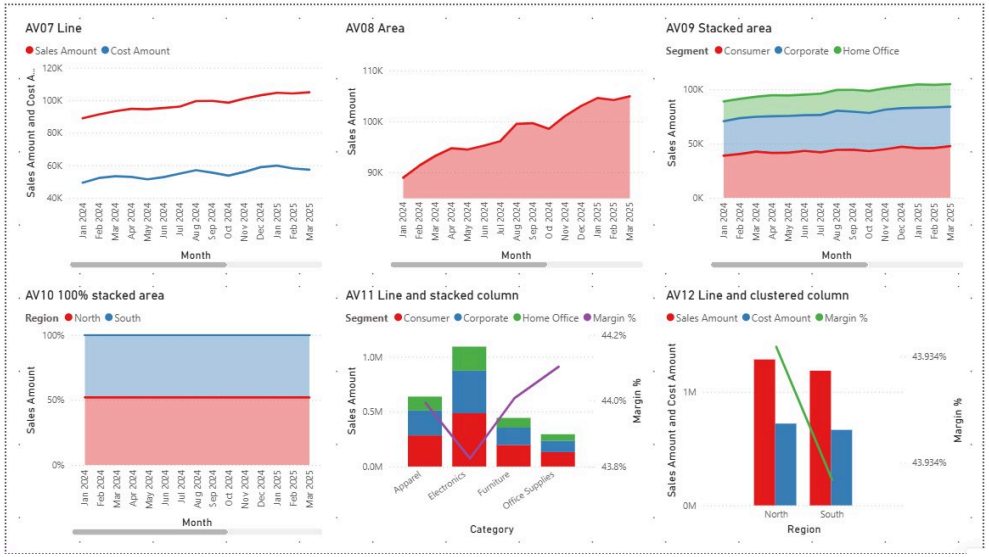
DIFFERENCES

- Excel's value axis uses 0.4M steps; Power BI shows 0M / 1M

05 All visuals 2 – Lines and areas MATCHES

EXCEL MODE · EXCEL CHARTS

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

- Line, area, stacked area and 100% stacked area as native charts over the 24 months
- Both combo charts: columns plus the Margin % line on a secondary axis

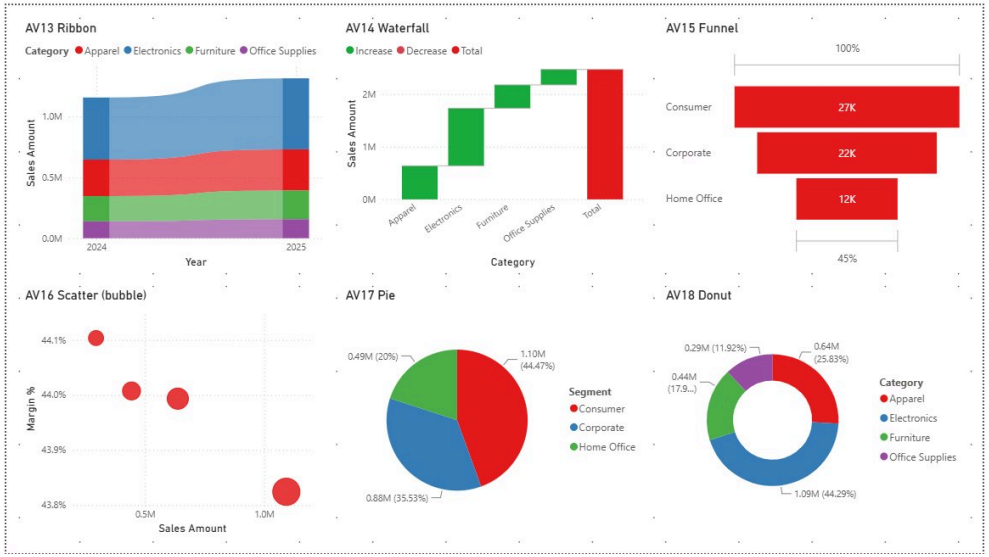
DIFFERENCES

- Power BI scrolls the month axis; Excel shows all 24 months at once
- AV12: Excel's line axis starts at 0%, so Margin % (43.93%–43.94%) looks flat where Power BI zooms in

All visuals 3 – Parts, flows and spread

CLOSE

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

- Waterfall: the same steps and total in the increase / total colours, labels like 2.47M
- Funnel: centred bars, largest on top, white value labels
- Scatter as a bubble chart (Size = Order Count), each bubble named; pie and donut with the same shares

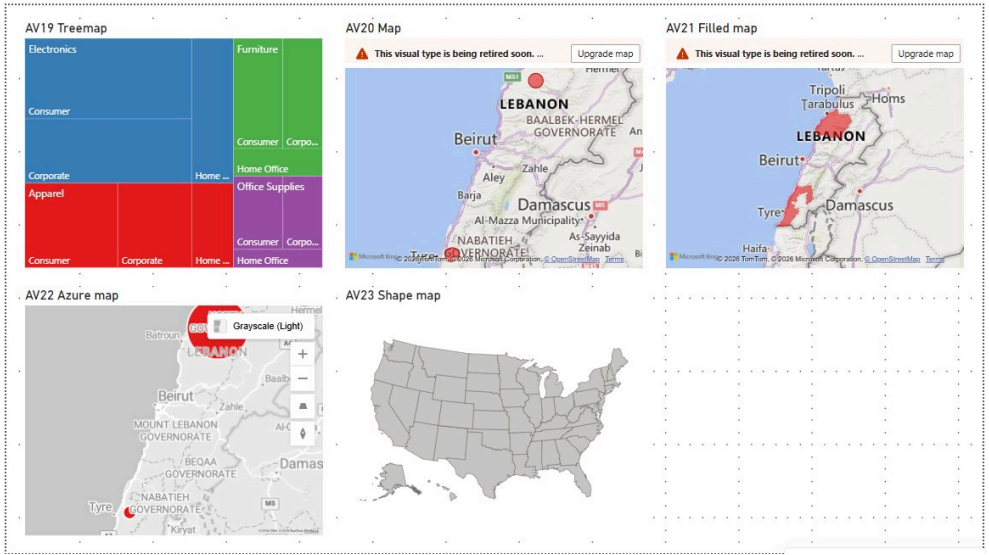
DIFFERENCES

- Ribbon chart is drawn as stacked columns: the ribbons between years aren't drawn
- The funnel's '100% / 45%' conversion rates aren't shown

07 All visuals 4 – Treemap and maps AS A TABLE

EXCEL MODE · DATA TABLES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel

DESIGN

All visuals 4 - Treemap and maps

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FILTERS No filters applied

AV19 Treemap

Colour by Category

Category	Segment	Sales Amount
Apparel	Consumer	284,099.72
Apparel	Corporate	226,576.89
Apparel	Home Office	127,175.43
Electronics	Consumer	486,209.16
Electronics	Corporate	388,041.14
Electronics	Home Office	219,366.96
Furniture	Consumer	196,668.89
Furniture	Corporate	158,096.82

12 rows · 4 grouped under [+] / [−] · expand before sorting

AV20 Map

Sales Amount

Region	Sales Amount
North	1,284,038.62
South	1,185,266.41

AV21 Filled map

Sales Amount

Region	Sales Amount
North	1,284,038.62
South	1,185,266.41

AV22 Azure map

Sales Amount

Region	Sales Amount
North	1,284,038.62
South	1,185,266.41

AV23 Shape map

Sales Amount

Region	Sales Amount
North	1,284,038.62
South	1,185,266.41

WHAT MATCHES

- Treemap tiles as a table with each category's cells in its tile colour
- Map, filled map, Azure map and shape map: locations and values, in the bubble colour

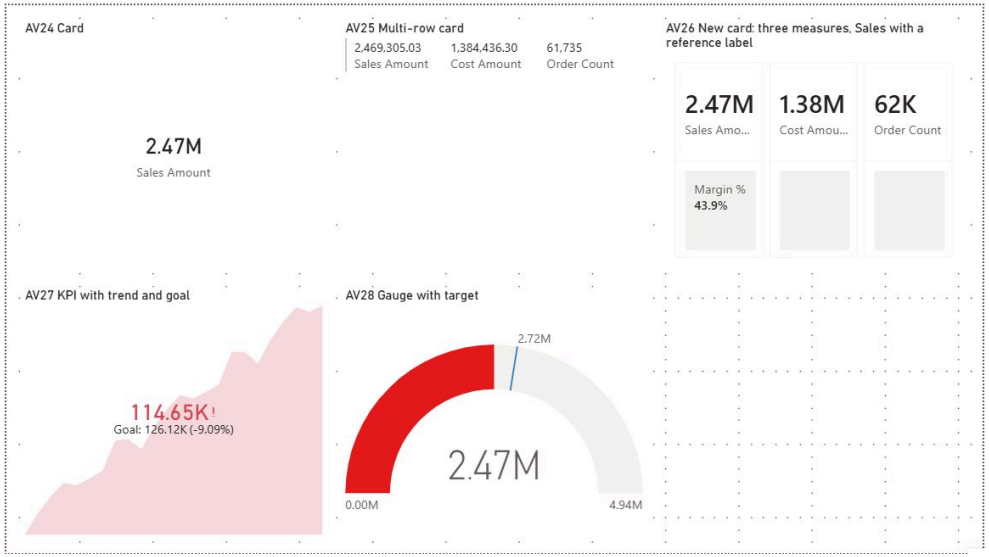
DIFFERENCES

- Excel has no map or treemap chart, so these stay tables in both modes

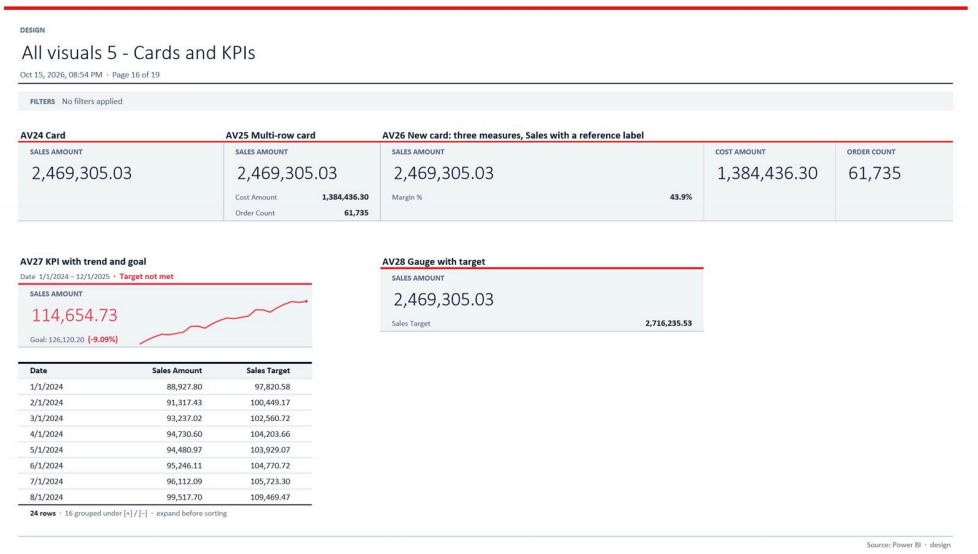
08 All visuals 5 – Cards and KPIs MATCHES

EXCEL MODE · DATA TABLES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



Date	Sales Amount	Sales Target
1/1/2024	88,927.80	97,820.58
2/1/2024	91,317.43	100,449.17
3/1/2024	93,237.02	102,560.72
4/1/2024	94,730.60	104,203.66
5/1/2024	94,480.97	103,929.07
6/1/2024	95,246.11	104,770.72
7/1/2024	96,112.09	105,723.30
8/1/2024	99,517.70	109,469.47

24 rows · 16 grouped under [+]/[−] · expand before sorting

WHAT MATCHES

- New card with three measures: three tiles side by side; its Margin % reference label (43.9%) under Sales
- KPI: 114,654.73 in the status colour, 'Goal: 126,120.20 (-9.09%)', 'Target not met', and an Excel sparkline of the trend
- Multi-row card and gauge as tiles with their other fields beneath

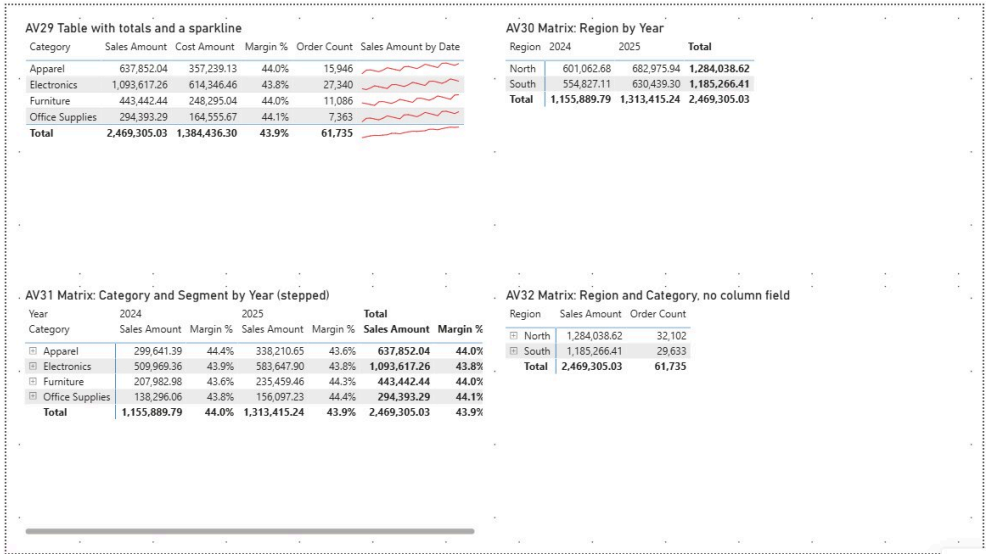
DIFFERENCES

- Power BI abbreviates (2.47M, 114.65K); Excel shows the exact values

09 All visuals 6 – Tables and matrices MATCHES

EXCEL MODE · DATA TABLES

POWER BI — Power BI Desktop



AV29 Table with totals and a sparkline

Category	Sales Amount	Cost Amount	Margin %	Order Count	Sales Amount by Date
Apparel	637,852.04	357,239.13	44.0%	15,946	
Electronics	1,093,617.26	614,346.46	43.8%	27,340	
Furniture	443,442.44	248,295.04	44.0%	11,086	
Office Supplies	294,393.29	164,555.67	44.1%	7,363	
Total	**2,469,305.03**	**1,384,436.30**	**43.9%**	**61,735**	

AV30 Matrix: Region by Year

Region	2024	2025	Total
North	601,062.68	682,975.94	**1,284,038.62**
South	554,827.11	630,439.30	**1,185,266.41**
Total	**1,155,889.79**	**1,313,415.24**	**2,469,305.03**

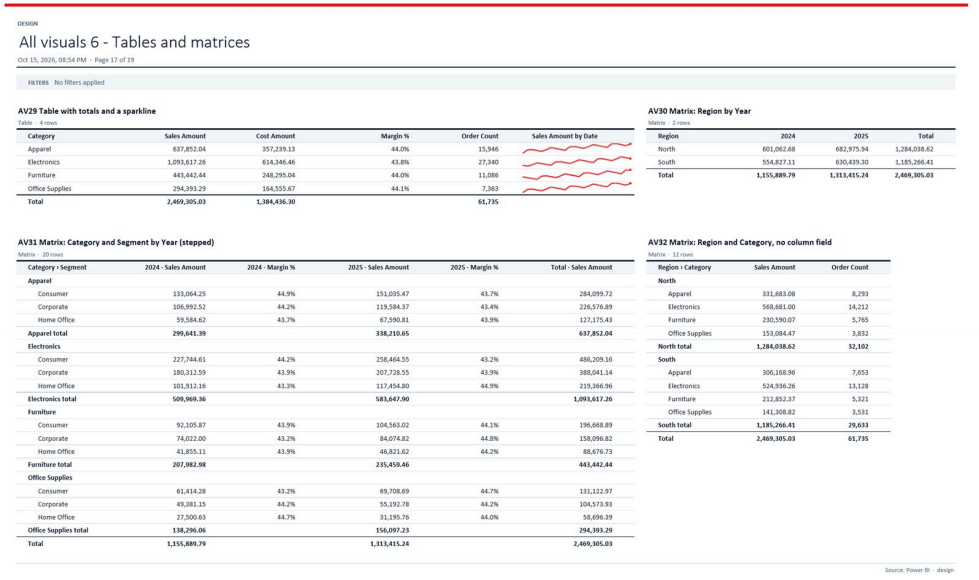
AV31 Matrix: Category and Segment by Year (stepped)

Year	2024		2025		Total	
Category	Sales Amount	Margin %	Sales Amount	Margin %	**Sales Amount**	**Margin %**
⊞ Apparel	299,641.39	44.4%	338,210.65	43.6%	**637,852.04**	**44.0%**
⊞ Electronics	509,969.36	43.9%	583,647.90	43.8%	**1,093,617.26**	**43.8%**
⊞ Furniture	207,982.98	43.6%	235,459.46	44.3%	**443,442.44**	**44.0%**
⊞ Office Supplies	138,296.06	43.8%	156,097.23	44.4%	**294,393.29**	**44.1%**
Total	**1,155,889.79**	**44.0%**	**1,313,415.24**	**43.9%**	**2,469,305.03**	**43.9%**

AV32 Matrix: Region and Category, no column field

Region	Sales Amount	Order Count
⊞ North	1,284,038.62	32,102
⊞ South	1,185,266.41	29,633
Total	**2,469,305.03**	**61,735**

EXCEL — Exported workbook, desktop Excel



DESIGN

All visuals 6 - Tables and matrices

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FILTERS No filters applied

AV29 Table with totals and a sparkline

Table · 4 rows

Category	Sales Amount	Cost Amount	Margin %	Order Count	Sales Amount by Date
Apparel	637,852.04	357,239.13	44.0%	15,946	
Electronics	1,093,617.26	614,346.46	43.8%	27,340	
Furniture	443,442.44	248,295.04	44.0%	11,086	
Office Supplies	294,393.29	164,555.67	44.1%	7,363	
Total	**2,469,305.03**	**1,384,436.30**		**61,735**	

AV30 Matrix: Region by Year

Matrix · 2 rows

Region	2024	2025	Total
North	601,062.68	682,975.94	1,284,038.62
South	554,827.11	630,439.30	1,185,266.41
Total	**1,155,889.79**	**1,313,415.24**	**2,469,305.03**

AV31 Matrix: Category and Segment by Year (stepped)

Matrix · 20 rows

Category › Segment	2024 - Sales Amount	2024 - Margin %	2025 - Sales Amount	2025 - Margin %	Total - Sales Amount
Apparel					
Consumer	133,064.25	44.9%	151,035.47	43.7%	284,099.72
Corporate	106,992.52	44.2%	119,584.37	43.4%	226,576.89
Home Office	59,584.62	43.7%	67,590.81	43.9%	127,175.43
Apparel total	**299,641.39**		**338,210.65**		**637,852.04**
Electronics					
Consumer	227,744.61	44.2%	258,464.55	43.2%	486,209.16
Corporate	180,312.59	43.9%	207,728.55	43.9%	388,041.14
Home Office	101,912.16	43.3%	117,454.80	44.9%	219,366.96
Electronics total	**509,969.36**		**583,647.90**		**1,093,617.26**
Furniture					
Consumer	92,105.87	43.9%	104,563.02	44.1%	196,668.89
Corporate	74,022.00	43.2%	84,074.82	44.8%	158,096.82
Home Office	41,855.11	43.9%	46,821.62	44.2%	88,676.73
Furniture total	**207,982.98**		**235,459.46**		**443,442.44**
Office Supplies					
Consumer	61,414.28	43.2%	69,708.69	44.7%	131,122.97
Corporate	49,381.15	44.2%	55,192.78	44.2%	104,573.93
Home Office	27,500.63	44.7%	31,195.76	44.0%	58,696.39
Office Supplies total	**138,296.06**		**156,097.23**		**294,393.29**
Total	**1,155,889.79**		**1,313,415.24**		**2,469,305.03**

AV32 Matrix: Region and Category, no column field

Matrix · 12 rows

Region › Category	Sales Amount	Order Count
North		
Apparel	331,683.08	8,293
Electronics	568,681.00	14,212
Furniture	230,590.07	5,765
Office Supplies	153,084.47	3,832
North total	**1,284,038.62**	**32,102**
South		
Apparel	306,168.96	7,653
Electronics	524,936.26	13,128
Furniture	212,852.37	5,321
Office Supplies	141,308.82	3,531
South total	**1,185,266.41**	**29,633**
Total	**2,469,305.03**	**61,735**

Source: Power BI · design

WHAT MATCHES

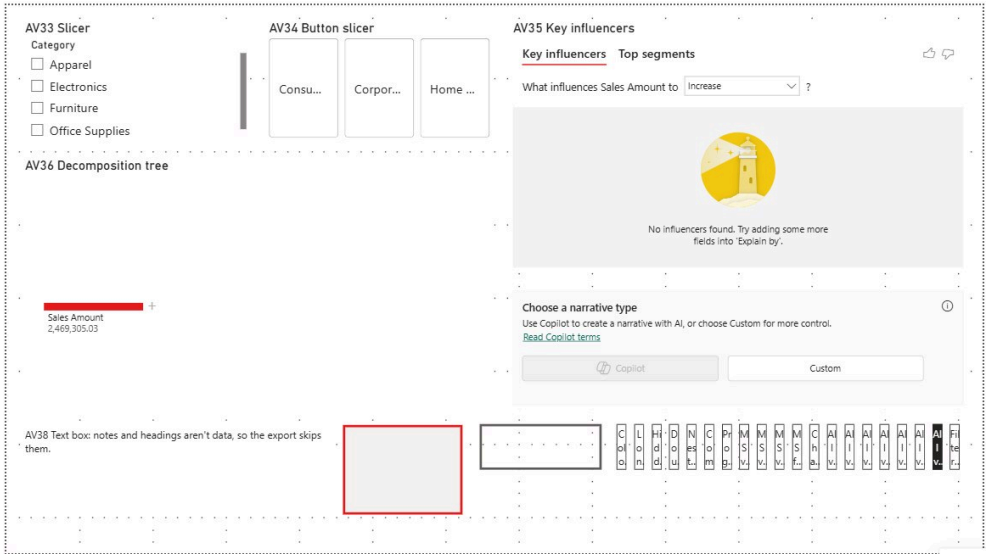
- Table: live totals row (SUBTOTAL) and the 'Sales Amount by Date' sparkline in every row
- Matrix Region × Year: one column per year and a live Total column
- Stepped matrices: group rows, indented members, group totals, a grand total, and Excel outline buttons

DIFFERENCES

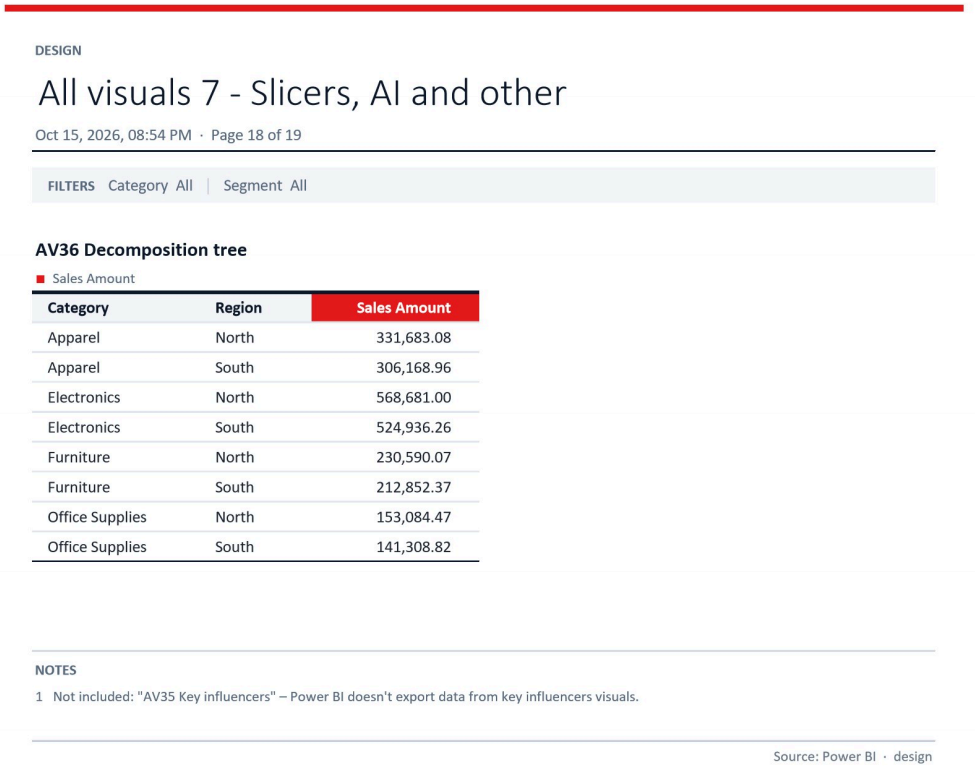
- Margin % totals stay blank: a ratio's total can't be added up from the rows (Power BI shows 43.9%)
- Power BI starts the stepped matrices folded; Excel opens them expanded

10 All visuals 7 – Slicers, AI and other MATCHES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



DESIGN

All visuals 7 - Slicers, AI and other

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FILTERS Category All | Segment All

AV36 Decomposition tree

Sales Amount

Category	Region	Sales Amount
Apparel	North	331,683.08
Apparel	South	306,168.96
Electronics	North	568,681.00
Electronics	South	524,936.26
Furniture	North	230,590.07
Furniture	South	212,852.37
Office Supplies	North	153,084.47
Office Supplies	South	141,308.82

NOTES

1 Not included: "AV35 Key influencers" – Power BI doesn't export data from key influencers visuals.

Source: Power BI · design

WHAT MATCHES

- Both slicers become chips in the filter bar ('Category All', 'Segment All')
- Decomposition tree: the analysed value by its explain-by fields
- Key influencers: named in the sheet's notes (Power BI doesn't export its data); text box, shapes, buttons and navigators left out quietly

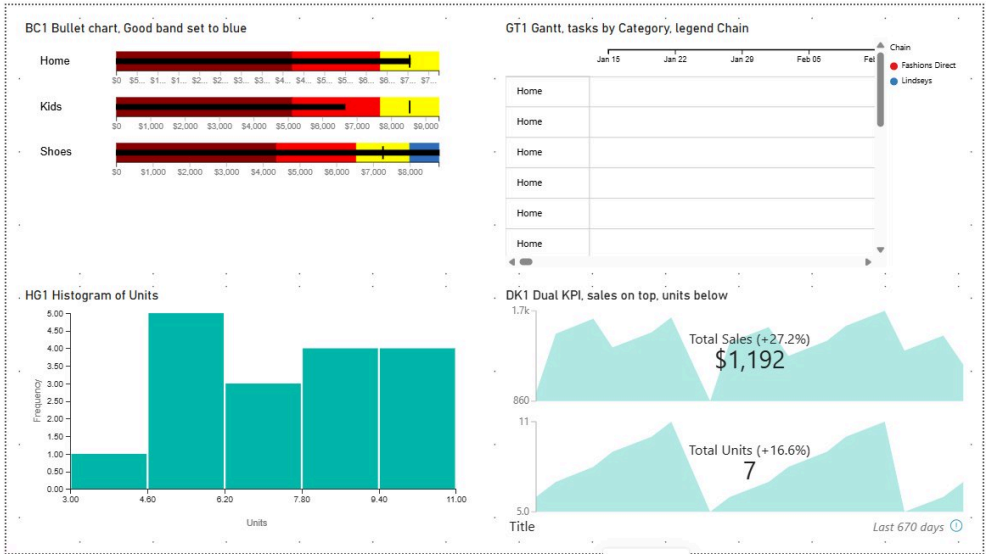
DIFFERENCES

- None that change the reading

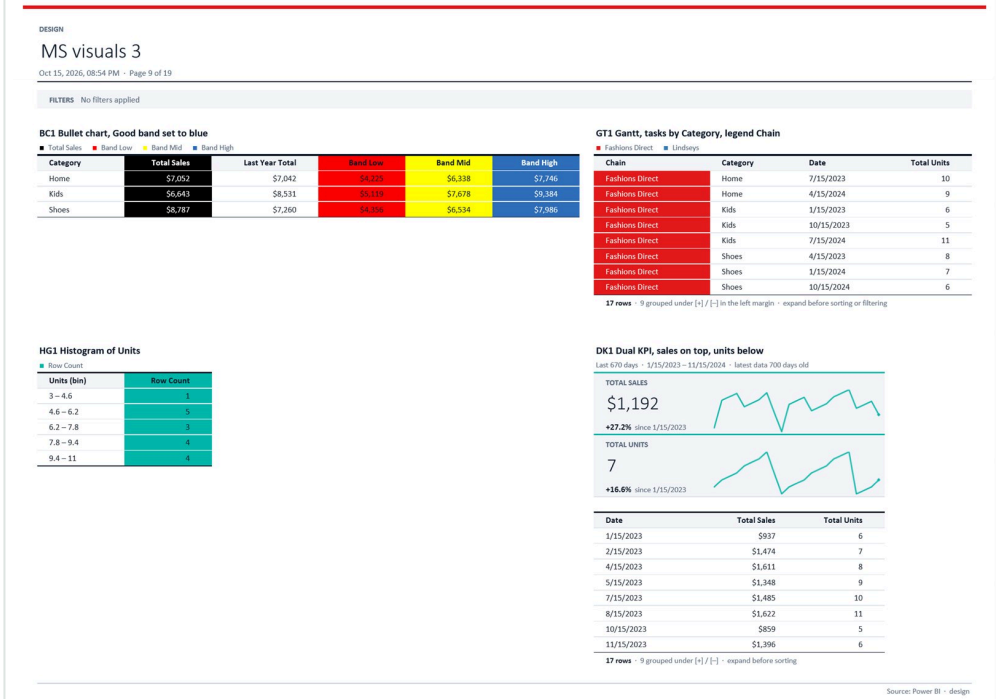
11 MS visuals 3 – Bullet, Gantt, Histogram, Dual KPI MATCHES

EXCEL MODE · DATA TABLES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



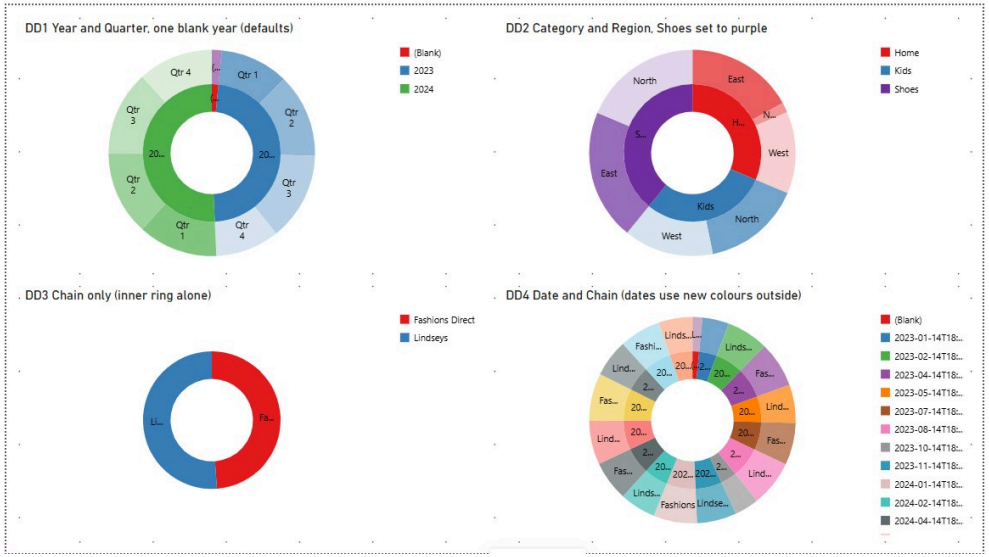
WHAT MATCHES

- Dual KPI: two trend cards, $1,192 (+27.2%) and 7 (+16.6%), 'Last 670 days', sparklines of the same shape
- Histogram: the visual's own bins (3–4.6 … 9.4–11) with counts 1, 5, 3, 4, 4
- Bullet chart band colours and the Gantt legend colours on the table

DIFFERENCES

- Bullet and Gantt charts have no Excel equivalent: tables in their colours

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



DESIGN

Double Donut

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FILTERS No filters applied

DD1 Year and Quarter, one blank year (defaults)

(Blank) 2023 2024

Year	Quarter	Total Sales
		$350
2023	Qtr 1	$2,411
2023	Qtr 2	$2,959
2023	Qtr 3	$3,107
2023	Qtr 4	$2,255
2024	Qtr 1	$2,803
2024	Qtr 2	$2,951
2024	Qtr 3	$2,999

9 rows · 1 grouped under [+] / [–] · expand before sorting

DD2 Category and Region, Shoes set to purple

Home Kids Shoes

Category	Region	Total Sales
Home	East	$3,810
Home	North	$350
Home	West	$2,892
Kids	North	$3,477
Kids	West	$3,166
Shoes	East	$4,599
Shoes	North	$4,188

DD3 Chain only (inner ring alone)

Fashions Direct Lindseys

Chain	Total Sales
Fashions Direct	$10,968
Lindseys	$11,514

DD4 Date and Chain (dates use new colours outside)

(Blank) 1/15/2023 2/15/2023 +14 more

Date	Chain	Total Sales
	Lindseys	$350
1/15/2023	Fashions Direct	$937
2/15/2023	Lindseys	$1,474
4/15/2023	Fashions Direct	$1,611

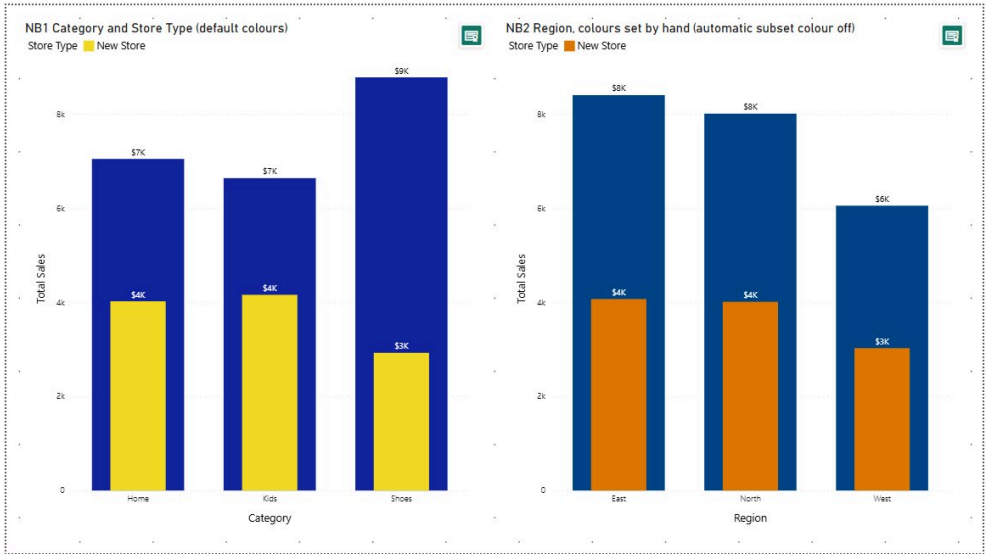
WHAT MATCHES

- Inner ring colours on each year (blank red, 2023 blue, 2024 green), outer ring shades on each quarter
- Shoes set to purple by hand is purple; the blank year reads (Blank) in the key

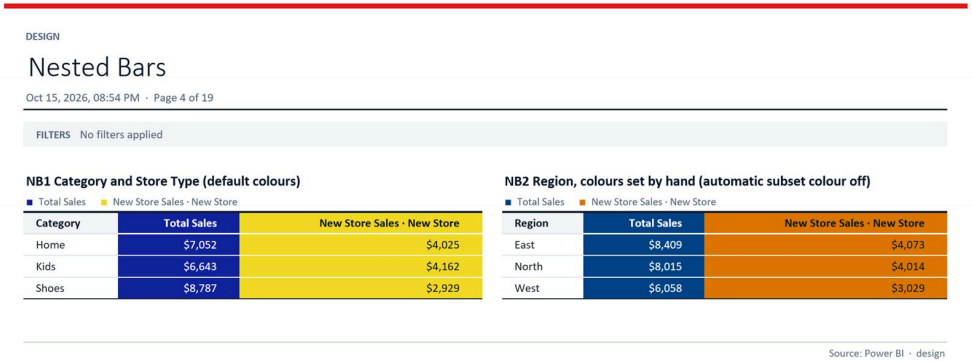
DIFFERENCES

- The rings are coloured cells; Excel has no double-ring chart

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



DESIGN

Nested Bars

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FILTERS No filters applied

NB1 Category and Store Type (default colours)

Total Sales · New Store Sales · New Store

Category	Total Sales	New Store Sales · New Store
Home	$7,052	$4,025
Kids	$6,643	$4,162
Shoes	$8,787	$2,929

NB2 Region, colours set by hand (automatic subset colour off)

Total Sales · New Store Sales · New Store

Region	Total Sales	New Store Sales · New Store
East	$8,409	$4,073
North	$8,015	$4,014
West	$6,058	$3,029

Source: Power BI · design

WHAT MATCHES

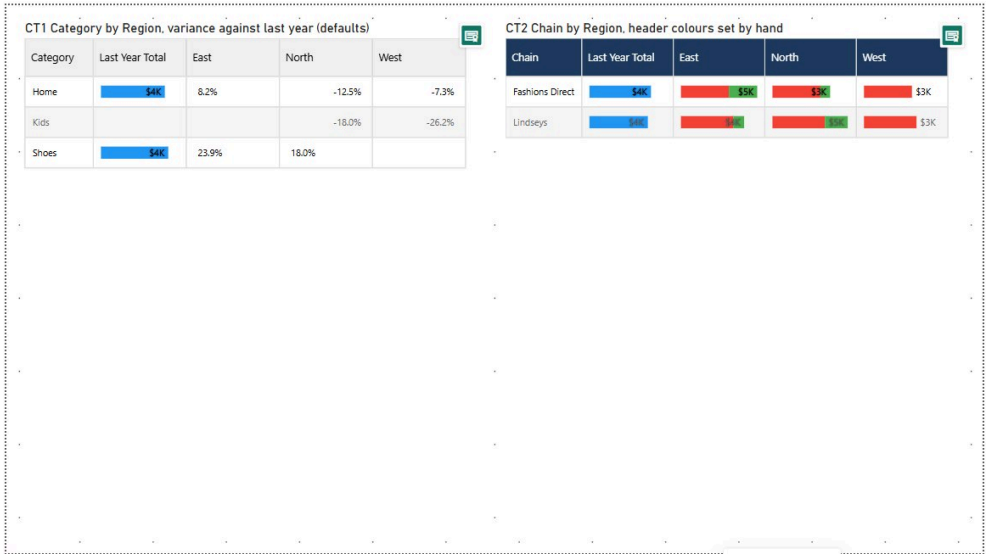
- A row per category: its total and the chosen subset's value
- Outer bar navy, inner bar its automatic complement, as in the visual

DIFFERENCES

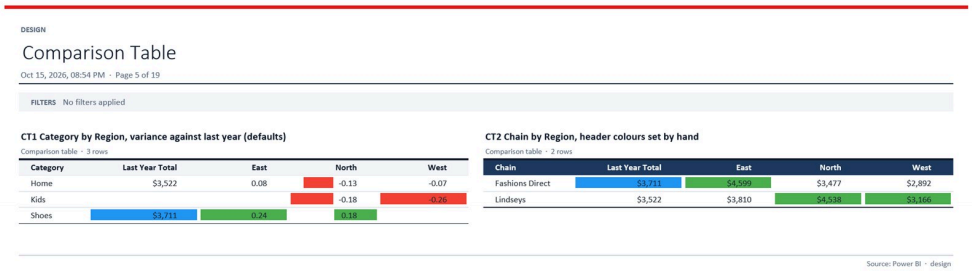
- None that change the reading

14 Comparison Table MATCHES

POWER BI — Power BI Desktop



EXCEL — Exported workbook, desktop Excel



WHAT MATCHES

- The visual's grid: a row per metric, the benchmark, a column per region
- Benchmark bars blue, value bars green and red for negatives, as Excel data bars

DIFFERENCES

- None that change the reading